

EXPLORATORY DATA ANALYSIS

Credit Risk Assessment — Financial Loan Portfolio

Case Study Report | 2,000,000 Loan Records

Dataset	9 CSV Files — 2,000,000 Loan Records
Domain	Banking & Credit Risk Analytics
Scope	EDA, Feature Engineering, OLS Regression Modelling
Tools	Python, Pandas, NumPy, Seaborn, Matplotlib, Statsmodels, Scikit-learn

1. INTRODUCTION

This report presents a comprehensive Exploratory Data Analysis (EDA) conducted on a financial loan portfolio comprising **2,000,000 loan records** sourced from nine interconnected datasets. The study focuses on understanding credit risk by examining borrower behaviour, loan characteristics, macroeconomic indicators, and historical default patterns.

Problem Statement: Financial institutions face the challenge of accurately identifying high-risk borrowers before loan sanction. This case study investigates the key variables driving loan default, estimates Loss Given Default (LGD), and explores the relationship between macroeconomic factors and portfolio-level credit losses.

Scope: The analysis covers data ingestion and quality treatment, missing-value classification, outlier handling, univariate and bivariate EDA, feature engineering, and OLS regression modelling with regularisation — culminating in five board-ready business recommendations.

2. OBJECTIVES OF THE ANALYSIS

- Assess the distribution and class imbalance of the `loan_status` variable (performing vs default).
 - Identify key numeric and categorical features that correlate with default probability and LGD.
 - Detect and treat data quality issues including negative values, impossible ranges, and duplicates.
 - Engineer meaningful features that capture repayment burden, bureau behaviour, and collateral strength.
 - Build and evaluate an OLS regression model for predicting LGD on defaulted loans.
 - Derive actionable business recommendations supported by statistical evidence.
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3. TOOLS AND TECHNOLOGIES USED

The entire analysis was performed in **Python 3** within a **Jupyter Notebook** environment. The table below summarises the libraries used at each stage of the workflow.

Library	Purpose
pandas / numpy	Data loading, merging, cleaning, transformation, and statistical computation.
matplotlib / seaborn	All visualisations — histograms, KDE plots, heatmaps, boxplots, line charts.
scipy.stats	Pearson correlation, independent-samples t-test (Welch's), probability plots.
statsmodels	OLS regression fitting, Durbin-Watson autocorrelation test, VIF computation.
scikit-learn	Ridge, Lasso, ElasticNet models; GridSearchCV; StandardScaler; train-test split.
fastparquet	Efficient columnar storage of downcasted DataFrames.

4. DATASET OVERVIEW

Nine CSV files were loaded and merged sequentially using left-joins on the primary key `loan_id`, resulting in a unified master DataFrame of **2,000,000 rows** and over 60 columns. Row-count assertions

after each join verified zero record loss, and an orphan-record report confirmed complete referential integrity across all child tables.

File	Description
loans_master.csv	Core loan attributes — amount, grade, interest rate, term, purpose.
loan_performance.csv	LGD, NPA flag, and payment performance metrics.
customer_bureau.csv	CIBIL score, delinquency history, revolving utilisation, credit history.
branch_region_economy.csv	State code, RBI repo rate, CPI inflation, GDP growth, COVID flag.
collateral_assets.csv	Collateral type and value.
credit_card_behavior.csv	Bank-card utilisation rates.
loan_enquiry_bureau.csv	Number of credit enquiries (30-day and 90-day windows).
monthly_emi_track.csv	Monthly EMI instalments.
payment_history.csv	Historical payment records.

5. DATA UNDERSTANDING AND PREPARATION

5.1 Data Quality Issues — 8 Defects Identified and Treated

A systematic quality screen flagged eight categories of defects, each treated with an appropriate strategy.

#	Column	Issue	Treatment
1	loan_amnt_inr	Negative loan amounts	Absolute value (sign flip)
2	int_rate_pct	Rate > 100% (decimal error)	Divide by 100
3	cibil_score	Outside 300–900 range	Clip to [300, 900]
4	annual_inc_inr	Negative income	Impute with median income
5	dti_pct	DTI > 100%	Cap at 99th percentile
6	emp_length_years	Negative employment tenure	Set to 0 (fresher)
7	revol_util_pct	Utilisation > 100%	Clip to 100
8	loan_id	Duplicate primary keys	Keep first occurrence

5.2 Missing Value Analysis (MCAR / MAR / MNAR)

Column	Classification	Imputation Method
mths_since_last_delinq	MNAR	Sentinel value 999 — absence means clean profile
mort_acc	MAR	Grouped median by home_ownership category
emp_length_years	MAR	Grouped median by emp_title; global median for residuals
il_util_pct	MCAR	Global median imputation

5.3 Outlier Treatment — Winsorisation

The six most skewed numeric columns were Winsorised at the 1st and 99th percentiles to suppress the influence of extreme values without discarding records. Log1p transformations were subsequently applied to `annual_inc_inr` and `loan_amnt_inr` (skew > 2.0) to satisfy the normality assumptions required by OLS regression.

6. EXPLORATORY DATA ANALYSIS

Q2a — Loan Status Distribution

Chart: Bar + Pie Chart

The dataset exhibits a severe class imbalance: ~96.12% performing loans vs ~3.88% defaults, yielding a majority-to-minority ratio of approximately 25:1. This imbalance means standard accuracy metrics are misleading; downstream models must use SMOTE or `class_weight='balanced'` to avoid learning a trivial majority-class predictor.

Q2b — CIBIL Score KDE by Loan Status

Chart: KDE Plot + Cohen's d

Performing borrowers have a mean CIBIL score approximately 21 points higher than defaulters. However, Cohen's d falls in the small-to-medium range, reflecting heavy distribution overlap. CIBIL alone is insufficient to separate defaulters reliably; it must be combined with other features.

Q2c — 12-Panel Histogram Grid

Chart: Histogram Grid (3x4)

Several features — particularly `annual_inc_inr` and `loan_amnt_inr` — exhibit skewness well above 2.0, confirming the need for log transformation before regression modelling. Features like `cibil_score` and `int_rate_pct` appear approximately normally distributed, requiring no transformation.

Q2d — Pearson Correlation Heatmap

Chart: Seaborn Heatmap

A strong positive correlation ($r > 0.75$) was detected between `open_acc` and `total_acc`, flagging multicollinearity. This pair must be handled via VIF-based feature selection before OLS regression. Most other numeric features showed weak-to-moderate pairwise correlations.

Q2e — Boxplots by Loan Status

Chart: Side-by-Side Boxplots

Boxplots of `int_rate_pct`, `dti_pct`, and `revol_util_pct` by loan status reveal that defaulted loans consistently have higher medians for all three variables. The IQR for defaulted loans is wider, indicating greater heterogeneity within the default cohort.

Q2f — Default Rate by Loan Grade

Chart: Bar Chart (A–G grades)	Default rates increase monotonically from Grade A (~1%) through Grade G (~25%). The sharpest single-step jump occurs between Grade C and Grade D (+9.4 pp), making this the critical risk-tier boundary. This validates the bank's grade-based risk-pricing framework.
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Q2g — Default Rate by Loan Purpose

Chart: Horizontal Bar Chart	Small business loans, car loans, and major purchases exhibit the highest default rates. Educational and home-improvement loans show the lowest rates. The highest-to-lowest default rate ratio is approximately 2.8x, justifying purpose-based risk differentiation.
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Q2h — Top 10 States by Default Rate

Chart: Bar Chart with Reference Lines	No single state breached the bank-wide average + 5 pp threshold in this dataset, suggesting that credit risk is driven more by borrower-level attributes than by regional macroeconomics. However, a small cluster of states consistently sits above the national average.
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Q2i — Annual Default Rate 2010–2024

Chart: Line Chart with COVID Shading	A visible spike in default rates occurred around 2020, attributable to the COVID-19 economic shock. The t-test on the covid_issue_year_flag confirms this increase is statistically significant ($t \approx 9.98$, $p < 0.05$), justifying inclusion of this flag as a model control variable.
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Q2j — RBI Repo Rate vs Default Rate

Chart: Dual-Axis Line Chart	Periods of aggressive rate tightening by the RBI correlate with subsequent increases in portfolio default rates, with an estimated 6–12-month transmission lag. Rising EMI burdens on floating-rate borrowers are the likely mechanism.
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Q2k — LGD Distribution

Chart: Histogram + KDE Overlay	The LGD distribution among defaulted loans is bimodal, with peaks near 0% and ~80%. This suggests two distinct recovery archetypes: near-complete recovery (collateralised loans) and near-total loss. The overall skewness is within an acceptable range, so direct log transformation is not mandatory.
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Q2l — CIBIL Score vs LGD Scatter

Chart: Scatter Plot + Regression Line	The Pearson correlation between CIBIL score and LGD among defaulted loans is approximately 0.004 — essentially zero. Once a borrower defaults, their credit score offers no predictive power for estimating loss magnitude, reinforcing the need for dedicated LGD features.
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7. DESCRIPTIVE STATISTICS

Key numeric features were profiled using standard descriptive statistics. Notable observations are summarised below:

Feature	Key Statistical Observation
cibil_score	Mean $\approx$ 650, roughly symmetric; scores cluster in the 600–750 band.
int_rate_pct	Right-skewed; a minority of loans carry very high rates ($>25\%$).
annual_inc_inr	Heavily right-skewed (skew $\gg 2$); log transformation required for modelling.
loan_amnt_inr	Right-skewed; large loans are outliers, treated via Winsorisation and log transform.
dti_pct	Most borrowers have DTI below 40%; very high DTI values ($>80\%$) were capped.
revol_util_pct	Wide spread; high utilisation borrowers are over-represented in the default cohort.
lgd_pct	Bimodal distribution peaking near 0% and $\sim 80\%$; mean LGD among defaults $\sim 50\%$.

8. FEATURE ENGINEERING

Ten new features were engineered across three categories to enrich predictive power:

Feature	Formula / Rationale
emi_to_income_ratio	$\text{installment_inr} \div (\text{monthly income})$ — measures repayment burden; strongest LGD predictor.
loan_to_income_ratio	$\text{loan_amnt_inr} \div \text{annual_inc_inr}$ — captures leverage relative to income.
rate_spread_pct	$\text{int_rate_pct} - \text{rbi_repo_rate_pct}$ — borrower's excess cost over policy rate.
real_interest_rate	$\text{int_rate_pct} - \text{cpi_inflation_pct}$ — inflation-adjusted cost of debt.
credit_util_composite	Weighted blend of <code>revol_util</code> , <code>bc_util</code> , <code>all_util</code> — holistic saturation score.
delinq_severity_score	$\text{delinq_2yrs} \times (1 + 1/\text{mths_since_last_delinq})$ — recency-weighted delinquency index.
enq_velocity_score	$4 \times \text{enquiries_30d} + \text{enquiries_90d}$ — credit-hunger / financial distress proxy.
income_stability_ratio	$\text{annual_inc_inr} \div (\text{emp_length} + 1)$ — normalises income by career tenure.
credit_depth_score	$\text{total_acc} \div (\text{credit_hist_years} + 1)$ — account-opening velocity.
collateral_coverage_ratio	$\text{collateral_value_inr} \div \text{loan_amnt_inr}$ — recovery buffer strength.

COVID Flag (Q3e): A binary indicator (`issue_year == 2020`) was created. An independent-samples Welch t-test confirmed significantly higher mean LGD for 2020-cohort loans ($t \approx 9.98$, $p < 0.05$), validating the flag as a critical structural control variable in regression models.

9. REGRESSION MODELLING AND DIAGNOSTICS

9.1 VIF-Based Feature Selection

Variance Inflation Factors were computed iteratively. Features with $VIF > 10$ (notably `rate_spread_pct`, which captures collinearity with `int_rate_pct`) were dropped in sequence until all remaining predictors satisfied $VIF \leq 10$, ensuring stable OLS coefficient estimates.

9.2 OLS Baseline Model

An Ordinary Least Squares model was fitted on the defaulted-loan subset with an 80/20 train-test split. The Durbin-Watson statistic near 2.0 confirmed no significant autocorrelation. However, the low R^2 value and the 4-panel diagnostic plots (Residuals vs Fitted, QQ Plot, Scale-Location, Cook's Distance) collectively revealed non-linearity, heteroskedasticity, and heavy-tailed residuals, indicating that a linear model is insufficient for LGD prediction.

9.3 Regularised Models — Ridge, Lasso, ElasticNet

GridSearchCV with 5-fold cross-validation was used to tune hyperparameters. All three regularised models produced near-zero Test R^2 , and Lasso shrunk most coefficients to zero — confirming that linear relationships between the selected features and LGD are negligible. Non-linear models (e.g., Gradient Boosting, Neural Networks) are recommended for production-grade LGD estimation.

10. KEY FINDINGS AND INSIGHTS

Class Imbalance (25:1)	Only 3.88% of loans defaulted. Standard accuracy is misleading; recall on the default class requires oversampling (SMOTE) or class-weight adjustments.
Grade Monotonicity	Default rates increase perfectly monotonically from Grade A to G. The C→D boundary (+9.4 pp) is the single largest risk step, making Grade a powerful ordinal risk signal.
CIBIL Score Limitation	While performing borrowers average 21 CIBIL points higher, the heavy overlap (small Cohen's d) means CIBIL alone cannot reliably screen defaulters. It must be combined with DTI and EMI ratios.
EMI-to-Income Ratio — Top Predictor	Among all engineered features, emi_to_income_ratio exhibited the strongest Pearson correlation with LGD. Borrowers with EMI/income > 50% show ~31% higher average loss severity.
Macroeconomic Sensitivity	The COVID-19 shock caused a statistically significant spike in LGD ($t \approx 9.98$). RBI repo rate increases correlate with default rate rises after a 6–12-month lag.
Collateral as Loss Buffer	Loans with a collateral coverage ratio > 1.2 show approximately 40% lower LGD, confirming that secured lending materially reduces loss exposure.
LGD Non-Linearity	OLS and regularised linear models failed to predict LGD (Test $R^2 \approx 0$), attributable to the bimodal LGD distribution and non-linear feature interactions. Tree-based ensemble models are indicated.

11. BUSINESS RECOMMENDATIONS

REC 1 — CIBIL Floor of 680	KDE analysis shows that borrowers with CIBIL < 650 default at 2.4× the portfolio average. Enforcing a hard floor of 680 (with override only for CCR > 1.5) could reduce NPA formation by an estimated 18–22%.
REC 2 — EMI-to-Income Cap at 50%	Borrowers with EMI/income > 50% exhibit ~31% higher LGD. Real-time income verification (GSTIN/ITR) and rejection of loans exceeding this threshold would materially reduce average loss severity.
REC 3 — Risk-Based Pricing for Grade D–G	The C→D default-rate jump of 9.4 pp is the largest in the grade ladder. A 150 bps risk premium on Grade D–G products would improve risk-adjusted returns without significantly reducing origination volume.
REC 4 — Sectoral Caps for High-Risk Purposes	Small business, car, and major-purchase loans default at 2.8× the rate of the safest purposes. Quarterly origination caps and front-line incentive restructuring can reduce segment-level NPA.
REC 5 — Collateral Mandate in High-Risk States	States persistently above the national default average should require collateral for loans > ₹5 Lakh. Loans with CCR > 1.2 show ~40% lower LGD, directly reducing net loss exposure in those geographies.

12. LEARNING OUTCOMES

- Applied a full EDA pipeline — from multi-file ingestion and data-quality remediation to feature engineering and regression modelling — on a realistic 2-million-record dataset.
- Classified missing values as MCAR, MAR, and MNAR, and selected context-appropriate imputation strategies.
- Used Winsorisation, log transformation, and VIF-based feature selection to prepare data for OLS regression.
- Gained practical understanding of class imbalance and its implications for model evaluation metrics.
- Interpreted Cohen's d, Pearson r, Durbin-Watson, and VIF statistics in a business context.
- Understood the limitations of linear models for predicting bimodal targets and the case for non-linear approaches.
- Translated statistical findings into quantified, evidence-backed business recommendations.

13. CONCLUSION

This case study delivered a thorough exploratory analysis of a 2,000,000-record credit portfolio. Nine datasets were merged, eight data-quality defects were identified and remediated, and missing values were classified and imputed using statistically appropriate methods.

EDA revealed a severe class imbalance (25:1 performing-to-default ratio), a monotonically ordered grade-based risk signal, and the practical limitation of CIBIL score as a standalone default predictor. Feature engineering produced ten new variables — led by `emi_to_income_ratio` as the strongest LGD correlate — and the COVID-19 flag was validated via Welch's t-test.

Regression modelling demonstrated that LGD prediction requires non-linear approaches; all linear models produced near-zero Test R^2 . Five business recommendations — covering credit floors, EMI caps, risk-based pricing, sectoral limits, and collateral mandates — were formulated with quantified impact estimates, providing an actionable framework for credit risk management.